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Mast30027 final exam 2016 solution

Modern Applied Statistics (University of Melbourne)



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1a) $\ell(p) = \log\binom{y+r-1}{r-1} + r\log(1-p) + y\log p$

$$\frac{\partial \ell}{\partial p} = \frac{-r}{1-p} + \frac{y}{p}$$

$$\frac{\partial^2 \ell}{\partial p^2} = \frac{-r}{(1-p)^2} - \frac{y}{p^2}$$

observed information is $\frac{r}{(1-p)^2} + \frac{y}{p^2}$

Fisher information is $I(p) = -\mathbb{E}\frac{\partial^2 \ell}{\partial p^2} = \frac{r}{(1-p)^2} \times \frac{r}{p(1-p)}$

$$= \frac{r}{1-p}\left(\frac{1}{1-p} + \frac{1}{p}\right) = \frac{r}{p(1-p)^2}$$

b) $p(n+1) = p(n) + \frac{\partial \ell}{\partial p}(p(n)) \Big/ I(p(n))$

$$p(1) = \frac{1}{2} + \left(\frac{-2}{1/2} + \frac{4}{1/2}\right) \Big/ \left(\frac{2}{1/8}\right)$$

$$= \frac{1}{2} + 4/16 = 3/4$$

$$p(2) = 3/4 + \left(\frac{-2}{1/4} + \frac{4}{3/4}\right) \Big/ \left(\frac{2}{3/64}\right)$$

$$= 3/4 + -8/5 \Big/ 128/3 = 11/16$$

c) MLE satisfies $\frac{r}{1-p} = \frac{y}{p} \equiv rp = y(1-p)$

$$\equiv p(r+y) = y \implies p = \frac{y}{r+y}$$

so $\hat{p} \approx N\left(\frac{y}{r+y}, \frac{p(1-p)^2}{r}\right)$

$$= N\left(\frac{2}{3}, \frac{2}{\sqrt{4}}\right)$$

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$$f(x)dx = P(x \le X < x+dx) = P(x^k \le X^k < (x+dx)^k)$$
$$= P(x^k \le X^k < x^k + kx^{k-1}dx) = f_Y(y) \underbrace{k x^{k-1} dx}_{\frac{dy}{dx}}$$

2. a) $f(x) = \exp\{\log k + (k+1)\log x + k\log\lambda - \lambda^k x^k\}$

$Y = X^k$ has pdf $g(y) = f(y^{1/k})\frac{dx}{dy}$

$$g(y) = k\, y^{(k-1)/k}\, \lambda^k e^{-\lambda^k y}\, \frac{1}{k} y^{1/k-1}$$
$$= \lambda^k e^{-\lambda^k y}$$

$Y \sim \exp(\lambda^k)$ which is from an exponential family.

b) The variance is of the form $\phi\mu^2$ but where ϕ is fixed (& depends on k). We could fit a model using gamma responses (which also have variance $\propto \mu^2$), but that would use an estimate of ϕ rather than the known value. However, fixing ϕ only effects the inference & not the fitted model, so we're OK. (and we could still do inference just by rescaling the s.e.s).

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3 a) model A $D = -2 \log L(\hat{\theta}) + 2 \log L(\text{full})$

$AIC = -2 \log L(\hat{\theta}) + 2p$

So $-2 \log L(\hat{\theta}) = 215.28 - 68 = 147.28$

$\Rightarrow \log L(\hat{\theta}) = -73.64$

$105.42 = -73.64 + 2 \log L(\text{full})$

$\Rightarrow \log L(\text{full}) = 89.53$

b) model B : AIC is smaller + χ^2 test rejects hypothesis that model A is correct

c) $Y_{ijk} \sim \text{Poisson}(\lambda_{ijk})$

where Y_{ijk} is the k-th obs with year = i + type = j

$$\log(\lambda_{ijk}) = \alpha + \beta_i + \gamma_j + \delta\sqrt{x_{ijk}} + (\beta\delta)_i \sqrt{x_{ijk}}$$

where x_{ijk} is service for Y_{ijk}

fitted values

α 0.193411

$\beta_{60} = 0$, $\beta_{65} = 0.203758$, etc

$\gamma_{ABE} = 0$, $\gamma_{CD} = -0.770174$

$\delta = 0.017483$

$(\beta\delta)_{60} = 0$, $(\beta\delta)_{65} = 0.002175$, etc

d) $\log(\hat{\lambda}) = 0.193411 + 1.304166 + (0.017483 + 0.001423)\sqrt{250}$

$= 1.7965$

$\Rightarrow \hat{\lambda} = 6.0286$

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3 e) Deviance is large and $\hat{\phi} \gg 1$

f) Poisson model assume $\text{Var } Y_i = \mathbb{E}\, Y_i$
Quasi-poisson assumes $\text{Var } Y_i = \phi\, \mathbb{E}\, Y_i$
for some ϕ.

g) $0.000215 \times \sqrt{2.829891} = 0.01046$

h) model C : F test fails to reject hypothesis that model C is correct.

i) $$F = \frac{(D_C - D_D)/(df_C - df_D)}{\hat{\phi}}$$

$$= \frac{15.142/5}{2.829891} = 1.7836 \quad \checkmark$$

df 3 + 31.

j) point 9 has high jackknife resid and v. high Cooks distance, so it fits poorly + has a big influence.

a poorly fitting point could also explain the poor fit of model B, so we would have to check model fit + $\hat{\phi}$ again.

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4 a)

model	D	df	95% sig
S + B + D	218.6	12 − 5 = 7	✓
S × B + D	36.4	12 − 5 − 2 = 5	✓
S + B × D	211.0	12 − 5 − 1 = 6	✓
S × D + B	182.4	12 − 5 − 2 = 5	✓
S × B + S × D	0.16	12 − 5 − 2 − 2 = 3	×
S × B + B × D	28.8	12 − 5 − 2 − 1 = 4	✓
S × D + B × D	174.8	12 − 5 − 2 − 1 = 4	✓
S × B + S × D + B × D	0.15	12 − 5 − 2 − 2 − 1 = 2	×

b) i) mutually dependent

ii) D indept. of S + B

iii) B + D conditionally indept given S

c) S × B + S × D significant (at 95% level)

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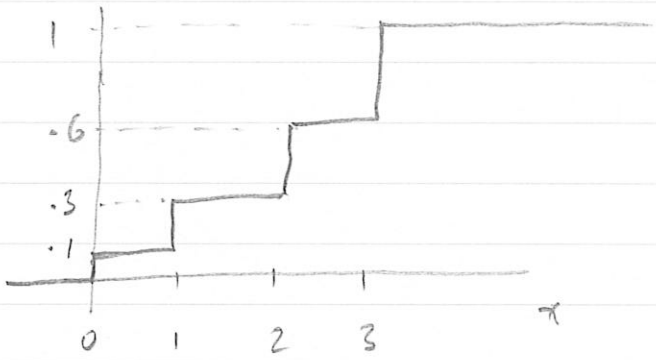
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5 a)

x	f1 $P(X=x)$	f2 $P(X=x)$
0	.1	1 − .9
1	.3 − .1	.9 − .7
2	.6 − .3	.7 − .4
3	1 − .6	.4

b)



c) f1 : $1 \times .1 + 2 \times .2 + 3 \times .7 = 2.5$

f2 : $1 \times .4 + 2 \times .3 + 3 \times .2 = 1.6$

f1 will run farther (on average)

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6 a) $P(X=x) = \int_0^1 \binom{m}{x} \theta^x (1-\theta)^{m-x} \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)} d\theta$

$= \frac{\binom{m}{x} B(\alpha+x, \beta+m-x)}{B(\alpha,\beta)}$

b) $p(k|y) \propto p(y|k)\, p(k)$

$\propto \binom{k}{y}\binom{N-k}{n-y}\binom{N}{k} B(\alpha+k, \beta+N-k)$

$= \frac{k!}{(k-y)!\, y!} \frac{(N-k)!}{(N-k-n+y)!\,(n-y)!} \frac{N!}{k!(N-k)!}$ — " —

$\propto \frac{1}{(k-y)!\,(N-n-(k-y))!} B(\alpha+k, \beta+N-k)$

$\propto \binom{N-n}{k-y} \frac{B(\alpha+y+(k-y), \beta+(n-y)+(N-n)-(k-y))}{B(\alpha+y, \beta+n-y)}$

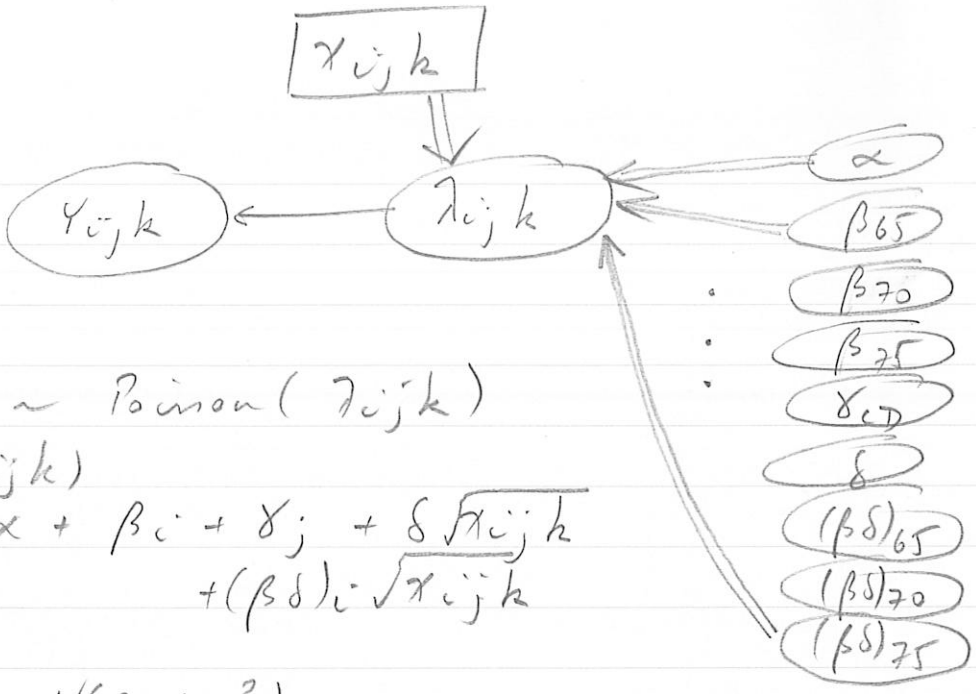
for $k \in \{y, \ldots, N-(n-y)\}$

so $k|y \sim y + \text{betabin}(N-n, \alpha+y, \beta+n-y)$

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7 a)



$Y_{ijk} \sim \text{Poisson}(\lambda_{ijk})$

$\log(\lambda_{ijk}) = \alpha + \beta_i + \gamma_j + \delta\sqrt{x_{ijk}} + (\beta\delta)_i\sqrt{x_{ijk}}$

$\alpha \sim N(0, 100^2)$

$\vdots$

$(\beta\delta)_{75} \sim N(0, 100^2)$

Y_{ijk} is k-th obs with year $= i$ & type $= j$

x_{ijk} is months service for Y_{ijk}

b) $Y_{ijk} \sim \text{negbin}(r, p_{ijk})$

$$p_{ijk} = \frac{r}{\mu_{ijk} + r}$$

$\log \mu_{ijk} = \cdots$

$r \sim \{1, 2, \ldots, 1000\}$

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8 a) given $\underline{x}^{(n)}$ propose $\underline{y}$ from $q(\underline{x}^{(n)}, \cdot)$

put $\alpha = \min\left\{1, \dfrac{\pi(\underline{y})\, q(\underline{y}, \underline{x}^{(n)})}{\pi(\underline{x}^{(n)})\, q(\underline{x}^{(n)}, \underline{y})}\right\}$

with prob α put $\underline{x}^{(n+1)} = \underline{y}$

o/w put $\underline{x}^{(n+1)} = \underline{x}^{(n)}$

b) let $p(\underline{x}, \underline{y})$ be (improper) trans. density for MH. then for $\underline{x} \neq \underline{y}$

$$\pi(\underline{x})\, p(\underline{x}, \underline{y}) = \pi(\underline{x})\, q(\underline{x}, \underline{y})\, \alpha$$

$$= \min\{\pi(\underline{x})\, q(\underline{x}, \underline{y}),\ \pi(\underline{y})\, q(\underline{y}, \underline{x})\}$$

$$= \pi(\underline{y})\, q(\underline{y}, \underline{x}) \min\left\{1, \frac{\pi(\underline{x})\, q(\underline{x}, \underline{y})}{\pi(\underline{y})\, q(\underline{y}, \underline{x})}\right\}$$

$$= \pi(\underline{y})\, p(\underline{y}, \underline{x}) \quad \text{as reqd.}$$

trivially true for $\underline{x} = \underline{y}$

c) π is limiting for MH (assuming irreducible + aperiodic). That is, if $\pi^{(n)}$ is dist at step n then $\pi^{(n)} \xrightarrow{d} \pi$

For small n $\pi^{(n)}$ may be a long way from π, so we get improved estimates if we drop early sample points.

The burn-in period is the no. of sample points we drop at the start, when the chain looks transitory rather than stationary.

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